

Address Resolution Protocol (ARP) & Internet Control Message Protocol (ICMP)

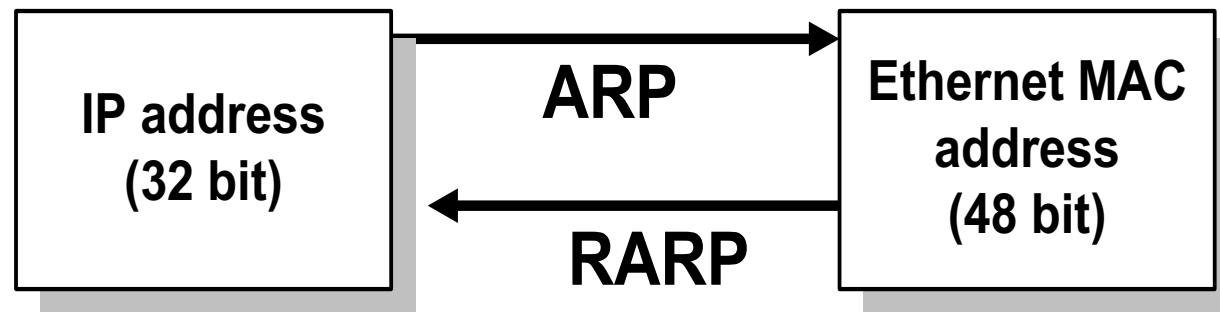
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Overview

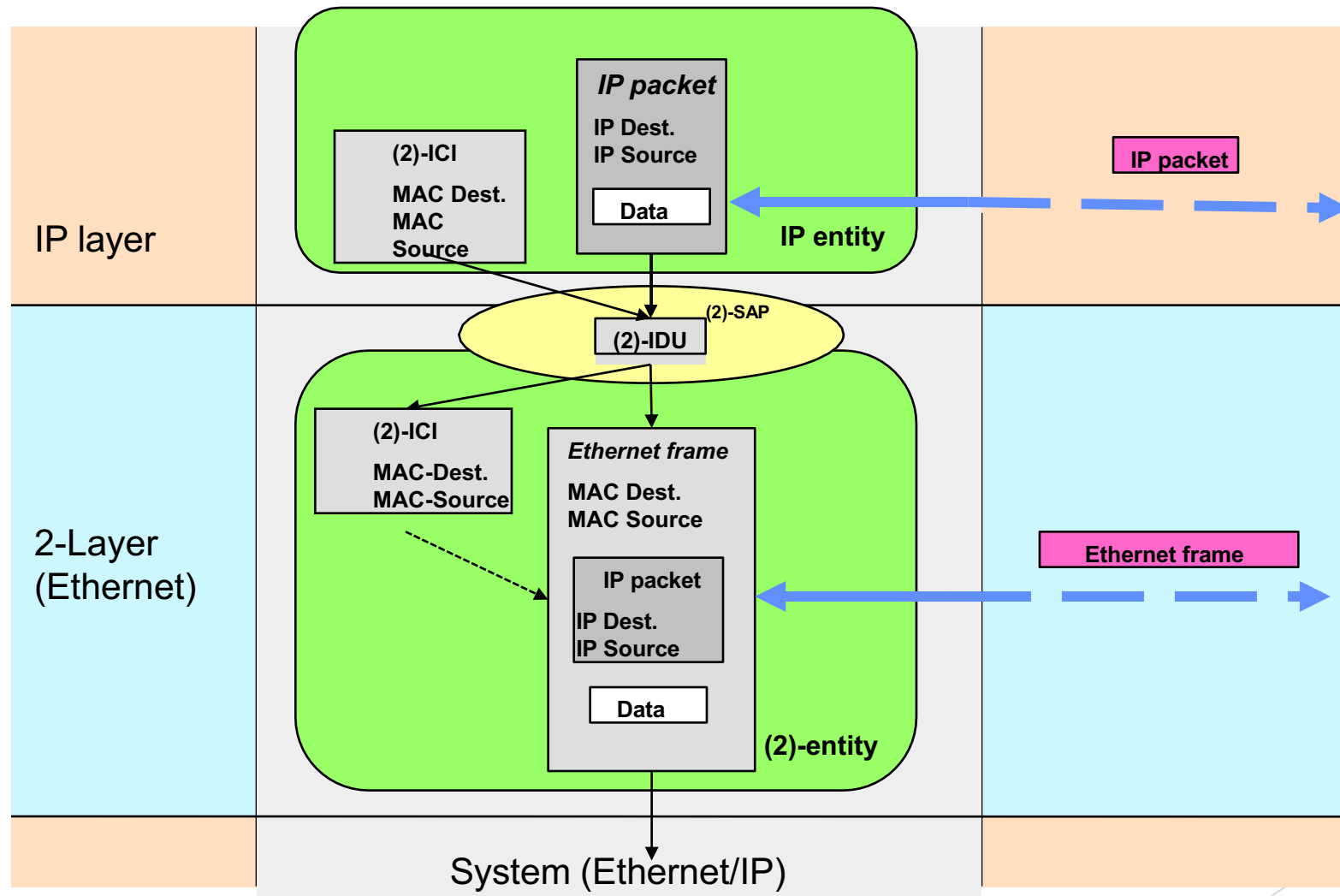
- ▶ The IP (Internet Protocol) relies on several other protocols to perform necessary control and routing functions:
 - ▶ Address resolutions (ARP)
 - ▶ Control functions (ICMP)
 - ▶ Multicast signaling (IGMP)
 - ▶ Setting up routing tables (RIP, OSPF, BGP, PIM, ...)

Address Resolution Protocol (ARP) and Reverse ARP (RARP)

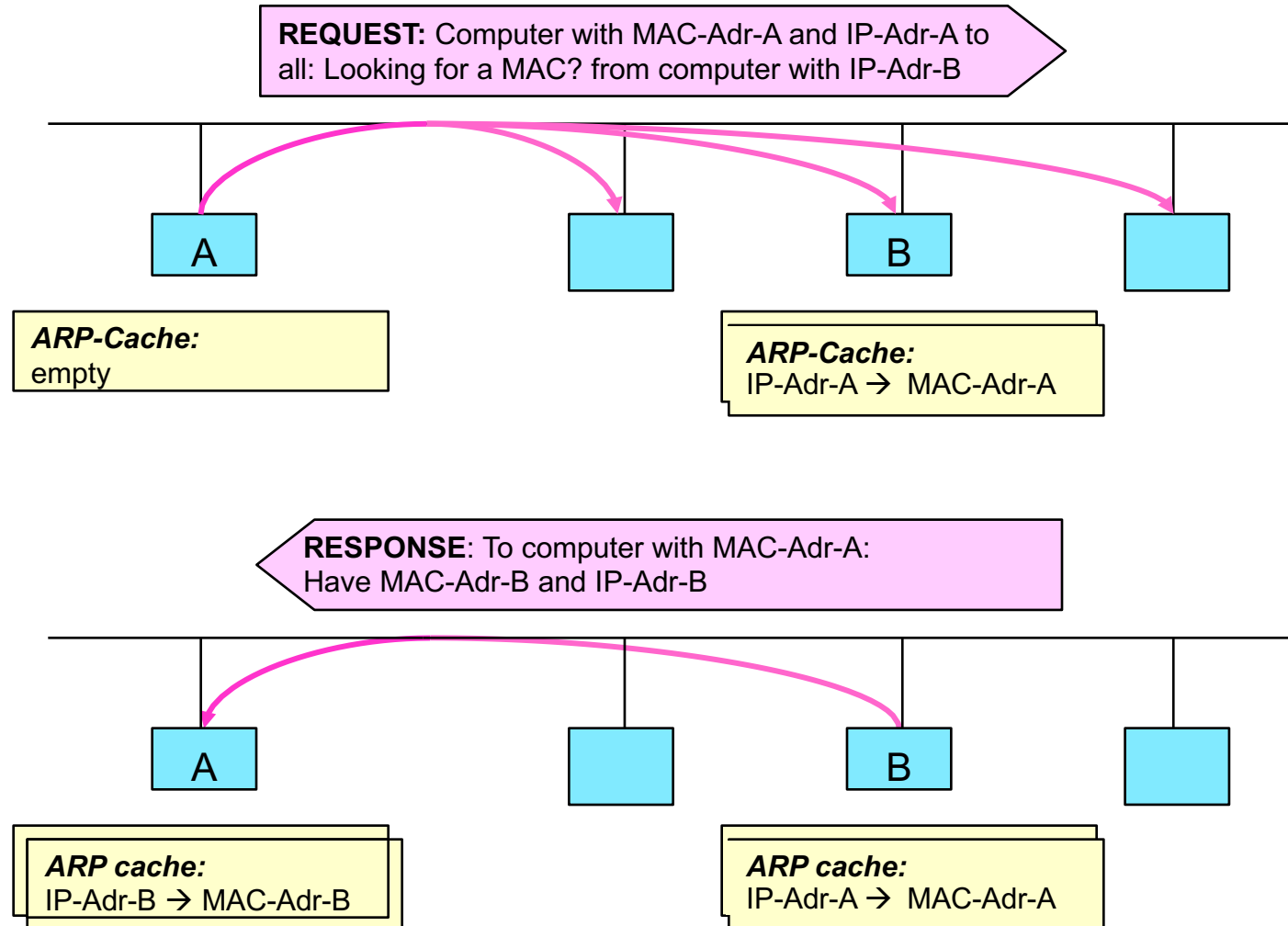
- ▶ Note:
 - ▶ The Internet is based on IP addresses
 - ▶ Data link protocols (Ethernet, FDDI, ATM) may have different (MAC) addresses
- ▶ The ARP and RARP protocols perform the **translation between IP addresses and MAC layer addresses**
- ▶ We will discuss ARP for broadcast LANs, particularly Ethernet LANs



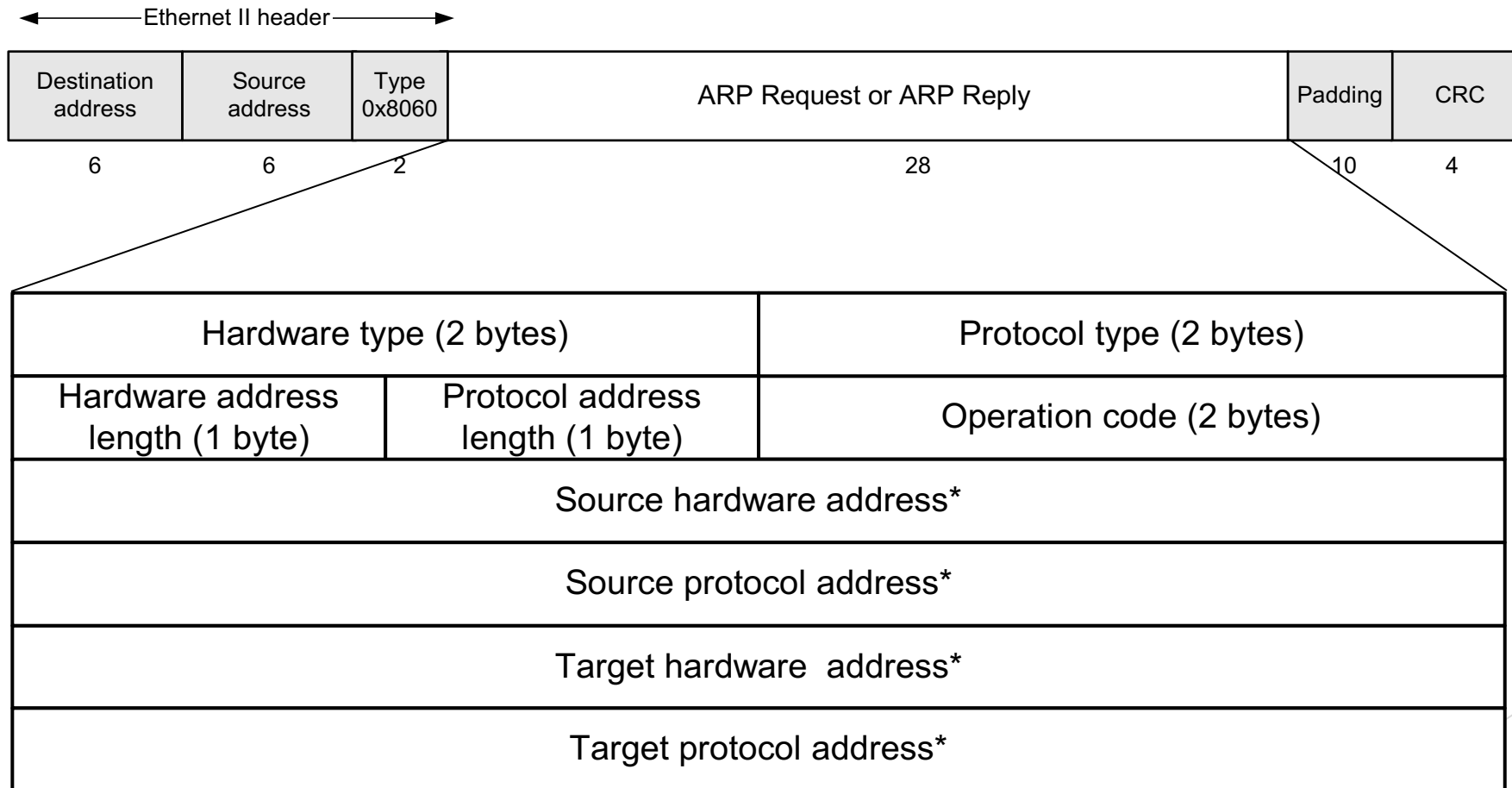
Coupling of the IP layer to the Ethernet layer according to OSI



ARP Protocol



ARP Packet Format (1)



* Note: The length of the address fields is determined by the corresponding address length fields

ARP Packet Format (2)

- ▶ **Hardware type (HTYPE)**
 - This field specifies the network link protocol type. Example: Ethernet is 1.
- ▶ **Protocol type (PTYPE)**
 - This field specifies the internetwork protocol for which the ARP request is intended. For IPv4, this has the value 0x0800. The permitted PTYPE values share a numbering space with those for [EtherType](#).
- ▶ **Hardware length (HLEN)**
 - Length (in [octets](#)) of a hardware address. Ethernet address length is 6.
- ▶ **Protocol length (PLEN)**
 - Length (in octets) of internetwork addresses. The internetwork protocol is specified in PTYPE. Example: IPv4 address length is 4.
- ▶ **Operation**
 - Specifies the operation that the sender is performing: 1 for request, 2 for reply.
- ▶ **Sender hardware address (SHA)**
 - Media address of the sender. In an ARP request this field is used to indicate the address of the host sending the request. In an ARP reply this field is used to indicate the address of the host that the request was looking for.
- ▶ **Sender protocol address (SPA)**
 - Internetwork address of the sender.
- ▶ **Target hardware address (THA)**
 - Media address of the intended receiver. In an ARP request this field is ignored. In an ARP reply this field is used to indicate the address of the host that originated the ARP request.
- ▶ **Target protocol address (TPA)**
 - Internetwork address of the intended receiver.

Example

► *ARP Request from Argon:*

Source hardware address: 00:a0:24:71:e4:44
Source protocol address: 128.143.137.144
Target hardware address: 00:00:00:00:00:00
Target protocol address: 128.143.137.1

► *ARP Reply from Router137:*

Source hardware address: 00:e0:f9:23:a8:20
Source protocol address: 128.143.137.1
Target hardware address: 00:a0:24:71:e4:44
Target protocol address: 128.143.137.144

ARP Cache

- ▶ Since sending an ARP request/reply for each IP datagram is inefficient, hosts maintain a cache (ARP Cache) of current entries. The entries expire after 20 minutes.
- ▶ Contents of the ARP Cache:
 - (128.143.71.37) at 00:10:4B:C5:D1:15 [ether] on eth0
 - (128.143.71.36) at 00:B0:D0:E1:17:D5 [ether] on eth0
 - (128.143.71.35) at 00:B0:D0:DE:70:E6 [ether] on eth0
 - (128.143.136.90) at 00:05:3C:06:27:35 [ether] on eth1
 - (128.143.71.34) at 00:B0:D0:E1:17:DB [ether] on eth0
 - (128.143.71.33) at 00:B0:D0:E1:17:DF [ether] on eth0

Things to know about ARP

- ▶ What happens if an ARP Request is made for a non-existing host?

Several ARP requests are made with increasing time intervals between requests. Eventually, ARP gives up.

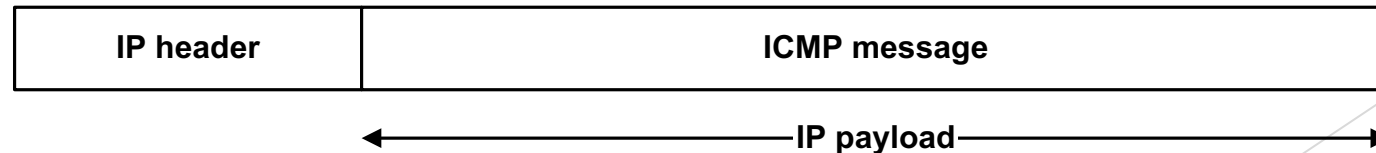
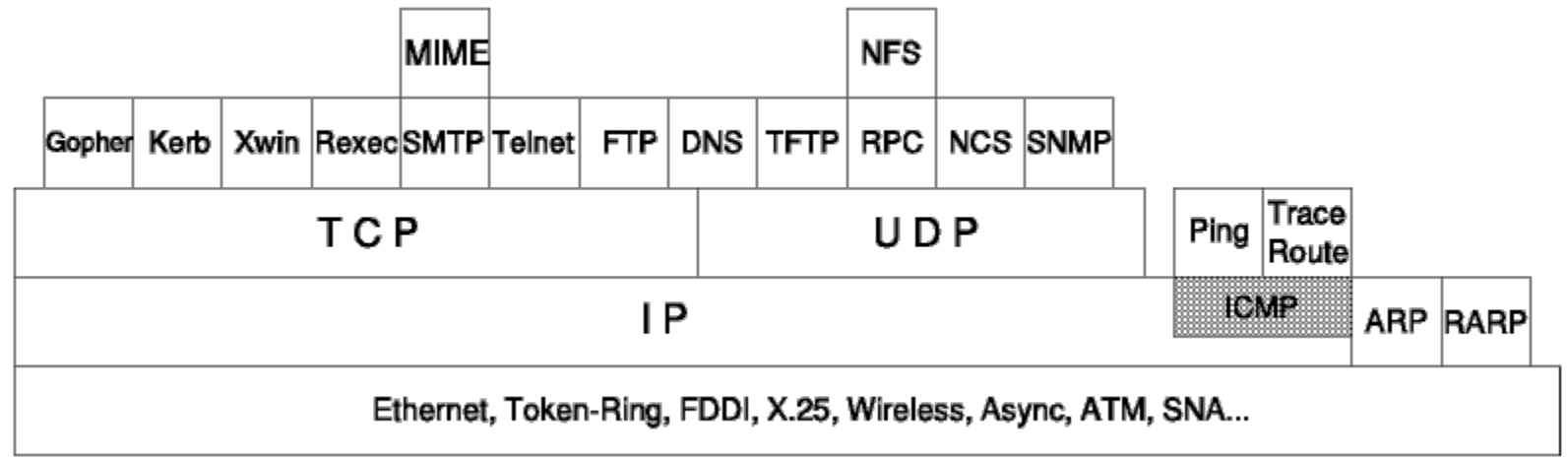
- ▶ What if a host sends an ARP request for its own IP address?

The other machines respond (**gratuitous ARP**) as if it was a normal ARP request.

This is useful for detecting if an IP address has already been assigned.

ICMP Overview

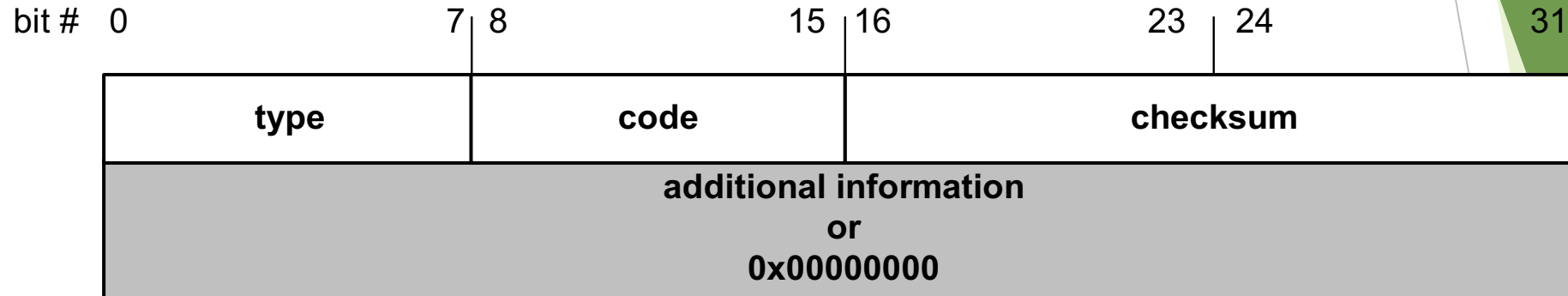
- ▶ The **Internet Control Message Protocol (ICMP)** is a helper protocol that supports IP with facility for
 - ▶ Error reporting
 - ▶ Simple queries



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- ▶ ICMP messages are encapsulated as IP datagrams

ICMP message format



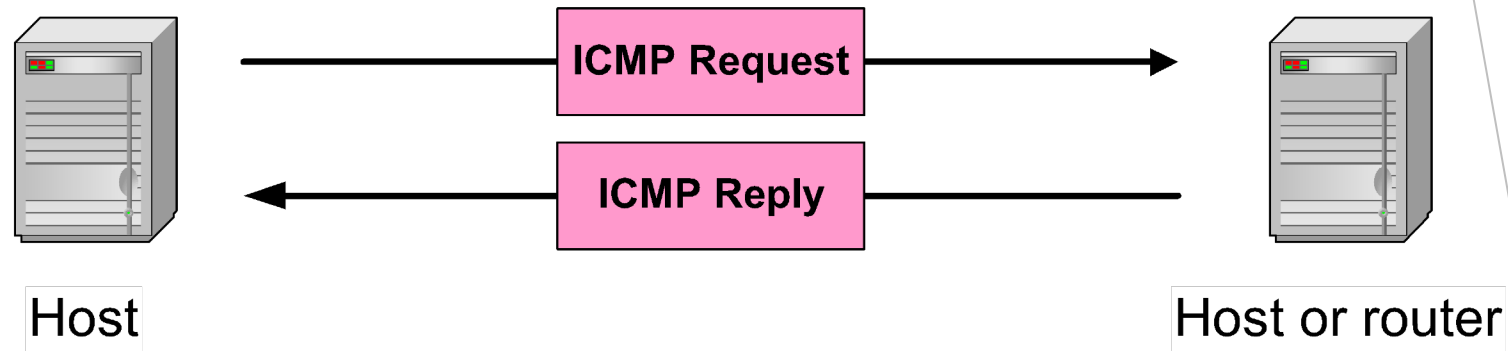
4 byte header:

- **Type (1 byte):** type of ICMP message
- **Code (1 byte):** subtype of ICMP message
- **Checksum (2 bytes):** similar to IP header checksum. Checksum is calculated over entire ICMP message

If there is no additional data, there are 4 bytes set to zero.

→ each ICMP messages is at least 8 bytes long

ICMP Query message

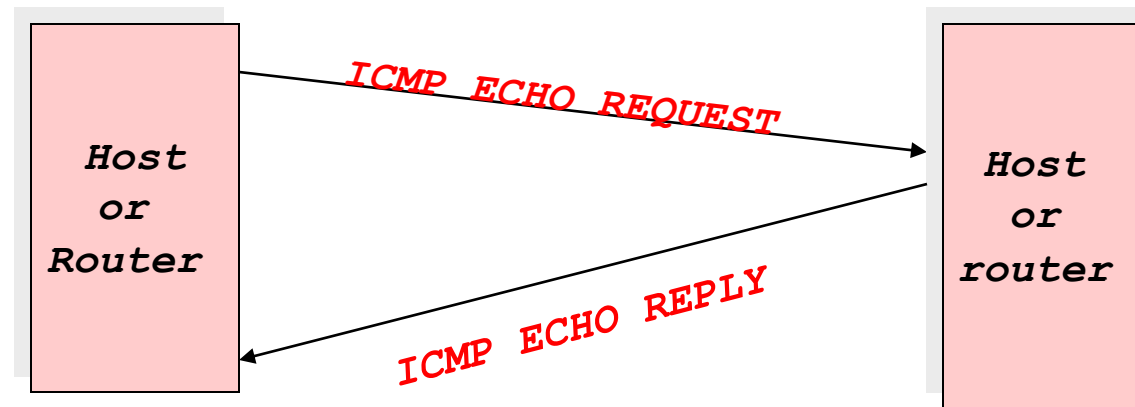


ICMP query:

- **Request** sent by host to a router or host
- **Reply** sent back to querying host

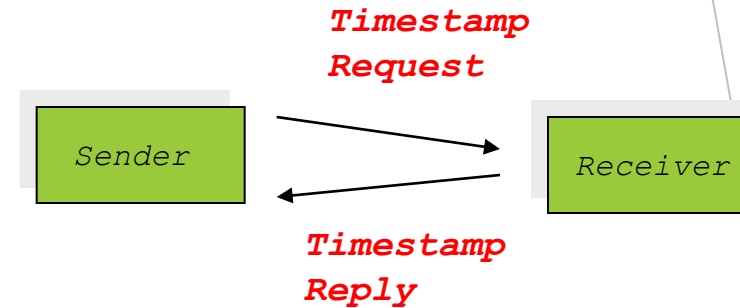
Example of a Query: Echo Request and Reply

- ▶ Ping's are handled directly by the kernel
- ▶ Each Ping is translated into an **ICMP Echo Request**
- ▶ The Ping'ed host responds with an **ICMP Echo Reply**



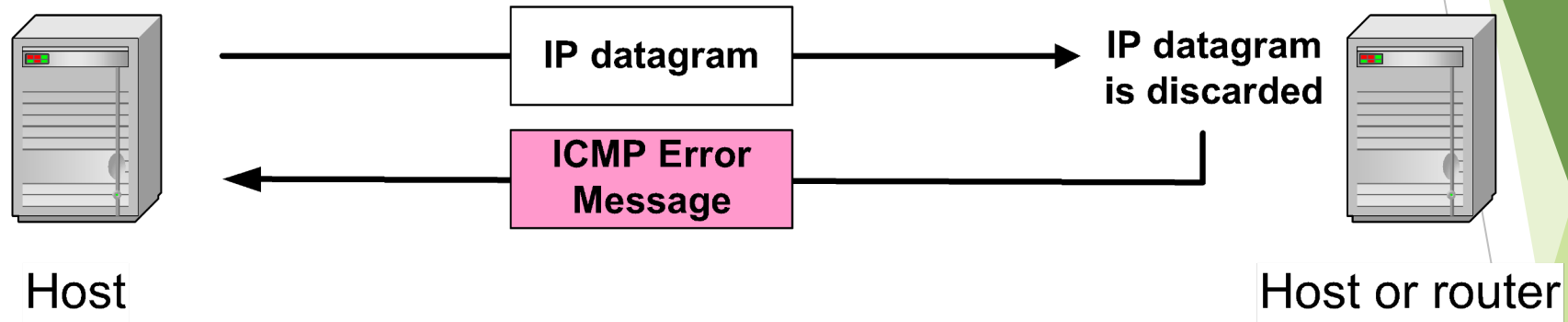
Example of a Query: ICMP Timestamp

- ▶ A system (host or router) asks another system for the current time.
- ▶ Time is measured in milliseconds after midnight UTC (Universal Coordinated Time) of the current day
- ▶ Sender sends a **request**, receiver responds with **reply**



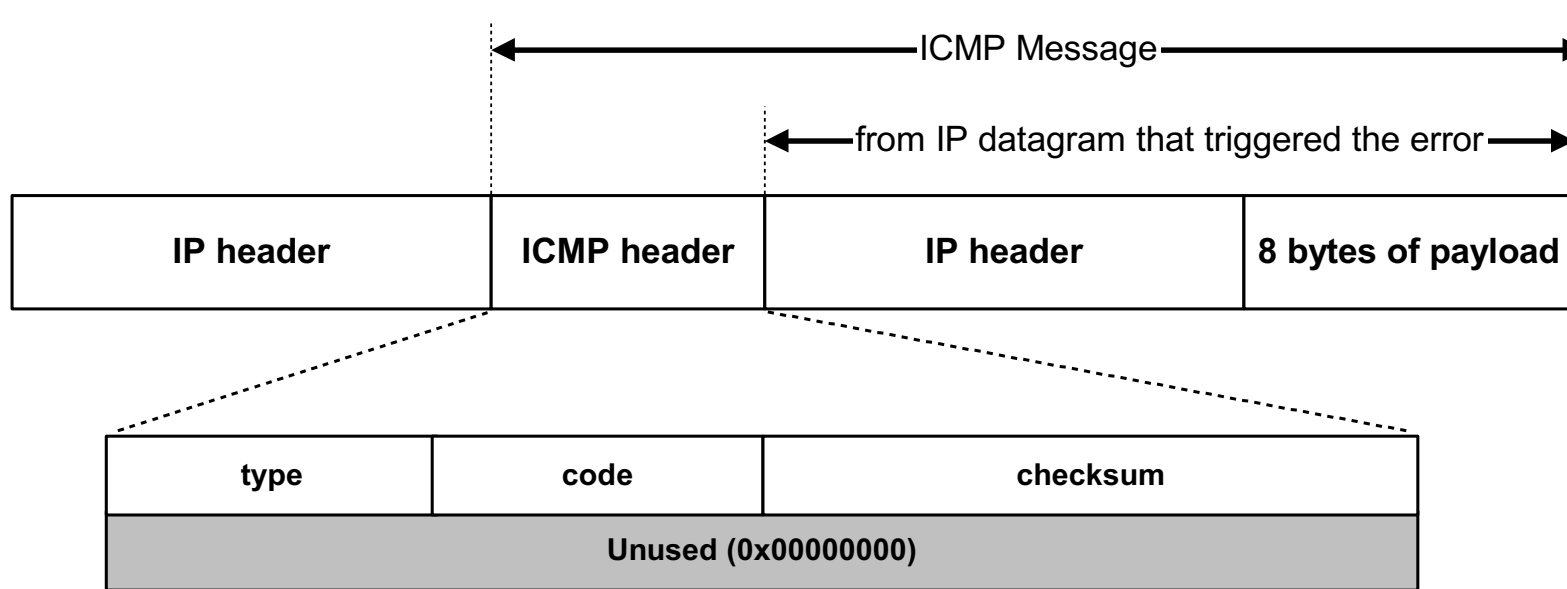
Type (= 17 or 18)	Code (=0)	Checksum
identifier		sequence number
32-bit sender timestamp		
32-bit receive timestamp		
32-bit transmit timestamp		

ICMP Error message



- ICMP error messages report error conditions
- Typically sent when a datagram is discarded
- Error message is often passed from ICMP to the application program

ICMP Error message



ICMP error messages include the complete IP header and the first 8 bytes of the payload (typically: UDP, TCP)

Frequent ICMP Error message

Type	Code	Description	
3	0–15	Destination unreachable	Notification that an IP datagram could not be forwarded and was dropped. The code field contains an explanation.
5	0–3	Redirect	Informs about an alternative route for the datagram and should result in a routing table update. The code field explains the reason for the route change.
11	0, 1	Time exceeded	Sent when the TTL field has reached zero (Code 0) or when there is a timeout for the reassembly of segments (Code 1)
12	0, 1	Parameter problem	Sent when the IP header is invalid (Code 0) or when an IP header option is missing (Code 1)

Some subtypes of the “Destination Unreachable”

Code	Description	Reason for Sending
0	Network Unreachable	No routing table entry is available for the destination network.
1	Host Unreachable	Destination host should be directly reachable, but does not respond to ARP Requests.
2	Protocol Unreachable	The protocol in the protocol field of the IP header is not supported at the destination.
3	Port Unreachable	The transport protocol at the destination host cannot pass the datagram to an application.
4	Fragmentation Needed and DF Bit Set	IP datagram must be fragmented, but the DF bit in the IP header is set.